Data Structures and Algorithms - 3 (25CS2103E)

Hospital Bed / Operating Room Scheduling Using Advanced Algorithms

Team Members

B. Bharath Sai Reddy — 2520090128

A. Chaitanya — 2520090092

Under the Esteemed Guidance of

Mrs. Ch. Anitha, Asst. Professor, CS&IT

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Abstract

- Efficient scheduling of hospital beds and operating rooms is a critical challenge because healthcare facilities have limited resources and multiple patient requirements.
- The proposed system develops an intelligent scheduling approach that assigns patients and surgical procedures to available beds, operating rooms and time slots while considering treatment duration, patient priority, staff availability, resource capacity and scheduling deadlines.
- The system models hospital scheduling as an optimization problem and applies network flow, assignment and approximation algorithms to generate high-quality schedules within practical execution time.
- The main objectives are to maximize resource utilization, reduce patient waiting time, minimize operating-room idle time and decrease overall completion time (makespan).
- The system also provides scheduling reports containing bed occupancy, operating-room utilization, scheduling efficiency and constraint violations.

Introduction

- Hospital beds and operating rooms are limited and costly resources.
- Multiple patients compete for the same rooms, beds, staff and time slots.
- Patients have different priorities, treatment durations and resource requirements.
- Staff and equipment availability further restrict feasible schedules.
- Emergency procedures may require immediate changes to an existing schedule.
- Surgery duration and patient requirements may be uncertain in real-world environments.
- Therefore, hospital scheduling is a complex resource-allocation and optimization problem.

Literature Survey

Al Amin, Baldacci & Kayvanfar (2025) — a comprehensive review of Operating Room Scheduling (ORS) research, 2000–2023

Key Findings

- OR scheduling involves multiple interacting resources and constraints — resource limitations, staff availability, patient variability, surgery duration, and emergency/urgent procedures.
- Research uses optimization methods under both deterministic and uncertain conditions.
- Common performance measures: patient waiting time, OR utilization, idle time, throughput, makespan and patient deferral.
- Heuristics and mathematical programming are frequently used for difficult linear and combinatorial optimization problems.

Problem Statement and Objectives

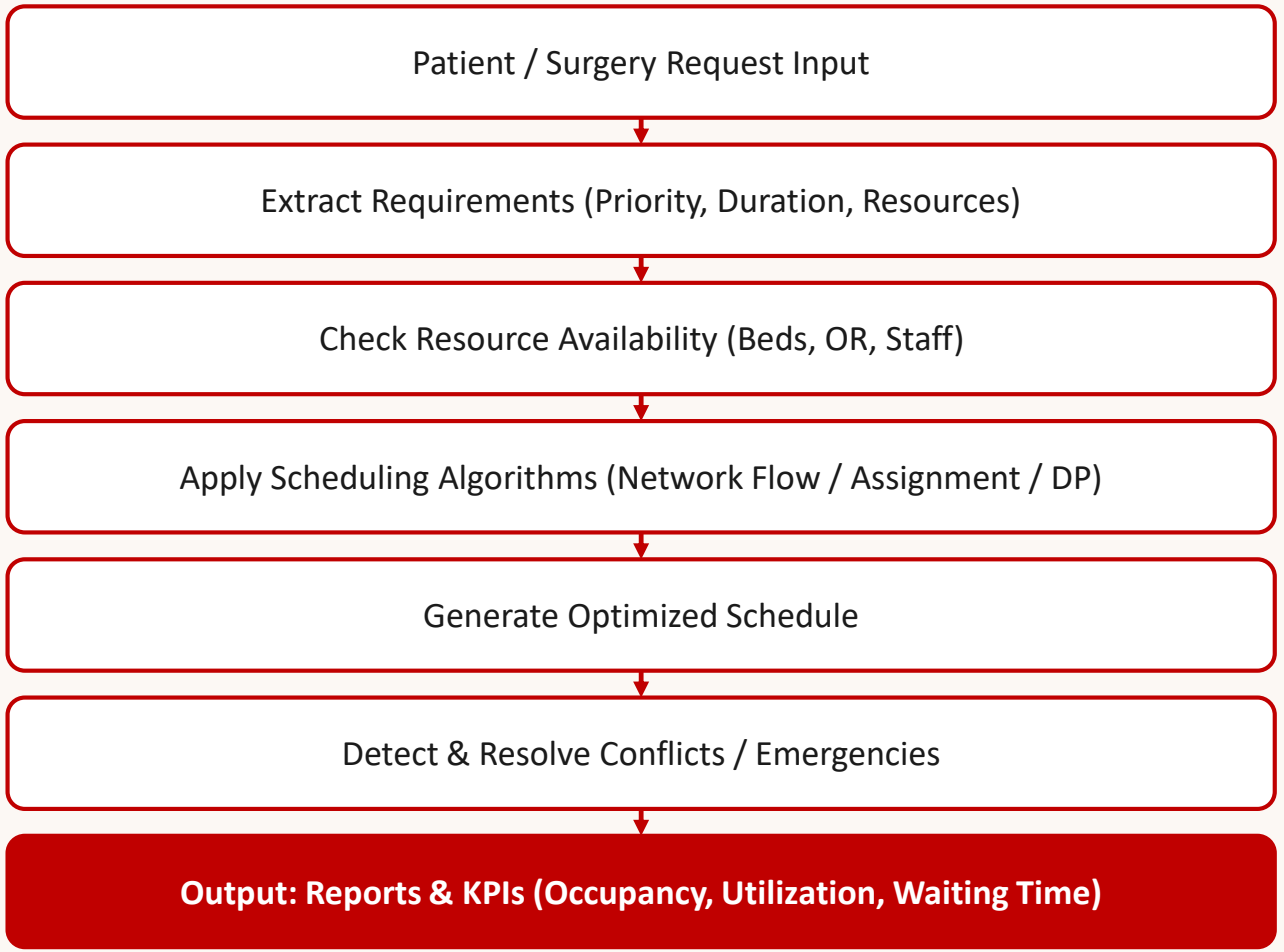
Problem Statement

- Hospitals must allocate patients and surgical procedures to limited beds, operating rooms, staff and time slots.
- Difficulty arises from limited resource capacity, variable surgery durations, and time-varying staff availability.
- Certain procedures require specific ORs or resources, and emergency cases can disrupt planned schedules.
- Growing patient volume causes an explosive growth in the number of possible schedules.
- Problem: how can beds and ORs be allocated efficiently while minimizing waiting time, idle time and completion time?

Objectives

- Maximize utilization of hospital resources.
- Minimize patient waiting time and scheduling conflicts.
- Prioritize urgent and high-priority patients.
- Consider staff and resource availability.
- Minimize overall completion time (makespan).
- Generate feasible schedules within practical execution time.
- Detect and report constraint violations; evaluate performance using measurable KPIs.

Flow Chart


Patient / Surgery Request Input
Extract Requirements (Priority, Duration, Resources)
Check Resource Availability (Beds, OR, Staff)
Apply Scheduling Algorithms (Network Flow / Assignment / DP)
Generate Optimized Schedule
Detect & Resolve Conflicts / Emergencies
Output: Reports & KPIs (Occupancy, Utilization, Waiting Time)

Results

Performance metrics tracked by the scheduling system

Metric	Desired Outcome
Patient Waiting Time	↓ Minimize
OR Idle Time	↓ Minimize
Resource Utilization	↑ Maximize
Scheduling Conflicts	↓ Minimize
Makespan	↓ Minimize
Throughput	↑ Maximize

Expected Output

- Patient-to-bed and patient-to-OR allocation
- Surgery time-slot and staff/resource allocation
- Patient priority handling
- Bed occupancy & OR utilization reports
- Scheduling efficiency & constraint-violation reports

Approach: Network flow and assignment algorithms allocate resources; dynamic programming optimizes time slots; approximation algorithms keep hard instances practical.

Hardware and Software Requirements

Hardware Requirements

- Computer / Laptop
- Minimum 4 GB RAM
- Dual-core processor or higher
- Minimum 5 GB available storage
- Keyboard and display

Software Requirements

- Operating System: Windows / Linux
- IDE: VS Code / IntelliJ IDEA
- Programming Language: Java
- Dataset: Patient, hospital resource and scheduling data
- Version Control: Git / GitHub
- Visualization: Charts / tables for performance analysis

Conclusion

- Hospital bed and operating-room scheduling is a complex, constraint-heavy optimization problem.
- Limited resources and multiple constraints make manual scheduling inefficient.
- The proposed system applies advanced DSA techniques to model and solve different parts of the problem.
- Network flow and assignment algorithms handle resource allocation.
- Dynamic programming supports time-slot optimization.
- Approximation algorithms provide practical solutions for computationally difficult scheduling instances.
- String algorithms support efficient information retrieval; randomized techniques explore alternative schedules.
- The system aims to improve resource utilization while reducing patient waiting time, OR idle time and makespan.

Future Scope

- Integrate real-time hospital data and dynamically update schedules when emergency cases arrive.
- Incorporate uncertainty in surgery duration; use historical data to predict duration.
- Integrate machine learning with optimization algorithms.
- Add real-time monitoring of beds and operating rooms.
- Extend the system to multiple hospitals.
- Include additional resources such as nurses, surgeons and specialized equipment.
- Explore advanced metaheuristic and adaptive scheduling approaches.
- Develop AI-assisted decision support for hospital administrators.

Thank You